

PATIENT MULTI-OMICS GENOMIC HEALTH SUMMARY

Patient Submitter ID: TCGA-44-3918 | Tumor: Lung Adenocarcinoma (LUAD) | Stage: Stage IIB

TCGA Project:	TCGA-LUAD	cBioPortal Study:	luad_tcga_pan_can_atlas_2018
Gender / Age:	male / 60 yrs	Vital Status / OS:	Deceased (38.5 mo)
Hybrid ML Risk:	79.3% (CRITICAL RISK)	VQC / Classical:	74.5% / 82.5%

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

Gene	Locus (GRCh38)	Transcript	Protein Change	ClinVar Status	COSMIC ID	gnomAD Pop AF	VAF %
EGFR	chr7:7p11.2	ENST00000275493	p.L858R Activating Kinase	Targetable (Osimertinib Sensitizing) VCV000016616	COSV51765492 FATHMM: 0.99 Pathogenic	0.000008	38.4%
KRAS	chr12:12p12.1	ENST00000311936	p.G12C GTPase Impairment	Targetable (Sotorasib) VCV000012582	COSV55546258 FATHMM: 0.98 Pathogenic	0.000004	34.2%

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ_{VQC} / R_Y / CNOT)

Gene Symbol	Canonical Ensembl ID	VAF Hotspot Frequency	Variational Angle θ (Radians)	Qubit Hilbert Space Projection
EGFR	ENSG00000146648	38.4%	$\theta = 1.9540 \text{ rad (112.0}^\circ\text{)}$	$ \psi\rangle = \cos(0.977) 0\rangle + \sin(0.977) 1\rangle$
KRAS	ENSG00000133703	34.2%	$\theta = 2.2150 \text{ rad (126.9}^\circ\text{)}$	$ \psi\rangle = \cos(1.107) 0\rangle + \sin(1.107) 1\rangle$

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:

First-line 3rd-generation EGFR TKI (Osimertinib) evaluation, brain MRI surveillance, and liquid biopsy monitoring for T790M.

Genomic Biomarker Summary: Identified somatic hotspot mutations in EGFR (p.L858R), KRAS (p.G12C). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the BlazeFinix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.